

Alexander Yoon

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Experience

Amazon

Vancouver, BC

Software Development Engineer Intern

May 2025 – Aug. 2025

- Designed and shipped **Overnight Upload** in Kotlin for Amazon Photos Android, delivering customer-facing production software for a product serving **200M+ customers**
- Architected asynchronous workflows across device state, charging conditions, network availability, and Android execution constraints, modeling a **35% improvement in successful backup completion**
- Built fault-tolerant client and cloud workflows with persistent state, retries, AWS S3, Lambda, and CloudWatch, decreasing failed upload recovery time by **40%**
- Owned feature delivery end to end across technical design, implementation, automated testing, staged rollout, monitoring, debugging, and post-launch iteration
- Collaborated with Product, QA, Android, and iOS engineers to evaluate tradeoffs, review code, resolve production issues, and improve system reliability

CIBC

Toronto, ON

Software Engineer Intern

Sep. 2024 – Apr. 2025

- Built production applications and backend services with **Java, Python, TypeScript, SQL, React, and REST APIs**, supporting customer and internal business workflows
- Designed scalable data and service pipelines processing **5M+ records** using PostgreSQL, PySpark, Databricks, Azure Data Factory, caching, and asynchronous processing
- Improved application and database performance through query optimization, indexing, parallel processing, and service-level profiling, reducing end-to-end runtime by **30%**
- Implemented automated testing, CI/CD, logging, monitoring, and integration validation, reducing recurring data and application inconsistencies by **25%**

Projects

CraveAI: Open-Source Full-Stack Software Platform

[GitHub](#) | [Live Demo](#)

- Built, open-sourced, and deployed a production full-stack application with **React, TypeScript, Python, FastAPI, PostgreSQL, and AWS**, owning frontend architecture, APIs, data models, and cloud deployment
- Designed REST APIs and service integrations with typed schemas, authentication, validation, retries, rate limiting, caching, and standardized error handling
- Built scalable product workflows across search, maps, recommendations, favorites, feedback, authentication, account management, and third-party integrations
- Improved recommendation **Precision@3 from 43.3% to 96.7%** while reducing unsupported outputs from **56.7% to 0%** through iterative system and ranking improvements
- Productionized the platform with CI/CD, observability, security controls, database migrations, failure recovery, and **114 automated backend and frontend tests**

Pocket AI: Cross-Platform Workflow Automation Platform

[GitHub](#)

- Led a team of **5 software engineers** in building a cross-platform application with React Native, TypeScript, Python, Flask, and REST APIs
- Designed end-to-end workflows spanning mobile UI, backend services, authentication, local persistence, notifications, synchronization, and external integrations
- Reduced average task-entry time by **60%**, from **45 seconds to 18 seconds**, while achieving **87% exact-field accuracy** across **100 noisy voice commands**
- Engineered retries, offline synchronization, conflict handling, and fault recovery, preserving **100% of records across 25 simulated network failures**

Education

McMaster University

Hamilton, ON

Bachelor of Software Engineering | GPA: 3.84/4.00

Graduated May 2026

Technical Skills

Languages: Python, Java, Kotlin, TypeScript, JavaScript, SQL, C, C++

Frontend & Mobile: React, React Native, Android, HTML, CSS, Tailwind, State Management

Backend & Data: FastAPI, Flask, Node.js, REST APIs, PostgreSQL, MongoDB, Redis, Databricks

Cloud & Systems: AWS, Azure, Docker, Distributed Systems, Caching, Queues, Concurrency, CI/CD

Engineering: Data Structures, Algorithms, System Design, Automated Testing, Observability, Git, Code Review